

Dimensionality Reduction

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Dimensionality Reduction

Definition:

Dimensionality Reduction is the process of transforming data from a high-dimensional space into a lower-dimensional space while preserving important information.

Given a dataset

$$X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix} \in \mathbb{R}^{n \times d}$$

where:

- n = number of samples
- d = number of features

Goal:

$$X \in \mathbb{R}^{n \times d} \rightarrow Z \in \mathbb{R}^{n \times k}$$

Why Dimensionality Reduction?

Challenges of High-Dimensional Data

- Increased computational cost.
- Storage requirements grow rapidly.
- Presence of redundant features.
- Risk of overfitting.
- Difficult visualization.

Curse of Dimensionality

As dimensionality increases:

$$\text{Volume} \propto r^d$$

where d is the number of dimensions.

Data become sparse, making learning more difficult.

Principal Component Analysis (PCA)

Idea:

Find new orthogonal directions that capture maximum variance.

Original variables:

$$(x_1, x_2, \dots, x_d)$$

are transformed into

$$(PC_1, PC_2, \dots, PC_d)$$

such that

$$\text{Var}(PC_1) > \text{Var}(PC_2) > \dots$$

Properties:

- Unsupervised technique.
- Components are orthogonal.
- Captures maximum information using fewer dimensions.

PCA: Data Centering

Given data matrix

$$X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix}$$

Compute feature mean:

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i$$

Center the data:

$$X_c = X - \mu$$

where each feature now has zero mean.

Centering ensures that PCA captures variance around the origin.

PCA: Covariance Matrix

The covariance matrix measures relationships between features.

$$C = \frac{1}{n-1} X_c^T X_c$$

where

$$C \in \mathbb{R}^{d \times d}$$

Each element:

$$C_{ij} = \text{Cov}(x_i, x_j)$$

Interpretation:

- Large positive value $\rightarrow$ positive correlation.
- Large negative value $\rightarrow$ negative correlation.
- Near zero $\rightarrow$ weak relationship.

PCA: Eigenvalue Decomposition

Compute eigenvectors and eigenvalues:

$$Cv = \lambda v$$

where

- v = eigenvector
- λ = eigenvalue

The eigenvectors define principal directions.

The eigenvalues indicate the amount of variance captured.

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$$

Select the first k eigenvectors:

$$W = [v_1, v_2, \dots, v_k]$$

PCA: Projection

Project data onto the lower-dimensional space:

$$Z = X_c W$$

where

$$W \in \mathbb{R}^{d \times k}$$

and

$$Z \in \mathbb{R}^{n \times k}$$

Example:

$$d = 100$$

features reduced to

$$k = 2$$

Variance Explained in PCA

The proportion of variance explained by component i :

$$\text{Explained Variance Ratio} = \frac{\lambda_i}{\sum_{j=1}^d \lambda_j}$$

Cumulative variance:

$$CV(k) = \frac{\sum_{i=1}^k \lambda_i}{\sum_{j=1}^d \lambda_j}$$

Choose the smallest k such that:

$$CV(k) \geq 95\%$$

PCA Example

Consider the dataset:

$$X = \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 4 & 3 \\ 5 & 4 \end{bmatrix}$$

Each row is a sample and each column is a feature.

Goal:

$$\mathbb{R}^2 \rightarrow \mathbb{R}^1$$

Reduce two features into one principal component while preserving maximum variance.

Step 1: Compute Mean

Feature means:

$$\mu_1 = \frac{2 + 3 + 4 + 5}{4} = 3.5$$

$$\mu_2 = \frac{1 + 2 + 3 + 4}{4} = 2.5$$

Mean vector:

$$\mu = \begin{bmatrix} 3.5 \\ 2.5 \end{bmatrix}$$

Step 3: Covariance Matrix

Covariance matrix:

$$\begin{aligned}C &= \frac{1}{n-1} X_c^T X_c \\&= \frac{1}{3} \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} \\&= \begin{bmatrix} 1.667 & 1.667 \\ 1.667 & 1.667 \end{bmatrix}\end{aligned}$$

Interpretation:

- Strong positive correlation.
- Both features vary together.

Step 4: Eigenvalues

$$|C - \lambda I| = 0$$

Eigenvalues:

$$\lambda_1 = 3.334$$

$$\lambda_2 = 0$$

The first principal component captures all variance.

The second component captures no variance.

Step 5: Eigenvector

Principal eigenvector:

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

This direction contains the maximum variance.

Step 6: Project Data

$$\begin{aligned} Z &= X_c v_1 \\ &= \begin{bmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} -2.121 \\ -0.707 \\ 0.707 \\ 2.121 \end{bmatrix} \end{aligned}$$

The original 2D data is now represented by a single feature.

PCA Result

Original Data:

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 4 & 3 \\ 5 & 4 \end{bmatrix}$$

Reduced Data:

$$\begin{bmatrix} -2.121 \\ -0.707 \\ 0.707 \\ 2.121 \end{bmatrix}$$

PCA successfully reduced:

2 dimensions $\rightarrow$ 1 dimension

while preserving nearly 100% of the variance.

Linear Discriminant Analysis (LDA)

Idea:

Find projections that maximize class separation.

Unlike PCA:

- PCA ignores class labels.
- LDA uses class labels.

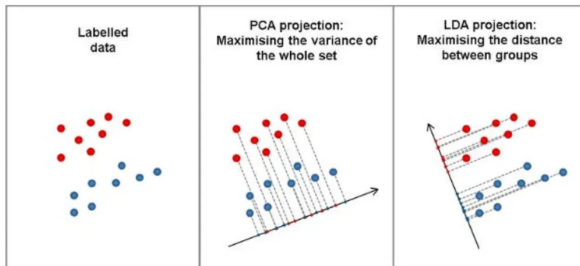
Goal:

$$\text{Maximize } \frac{\text{Between-Class Variance}}{\text{Within-Class Variance}}$$

Result:

- Better discrimination among classes.
- Useful for classification tasks.

Linear Discriminant Analysis



- PCA projects data along directions of maximum variance.
- LDA projects data along directions that maximize class separation.
- PCA ignores labels; LDA uses labels.

LDA: Within-Class Scatter Matrix

Suppose there are c classes.

For class i :

$$S_i = \sum_{x \in C_i} (x - \mu_i)(x - \mu_i)^T$$

where

$$\mu_i = \frac{1}{N_i} \sum_{x \in C_i} x$$

is the class mean.

Within-class scatter:

$$S_W = \sum_{i=1}^c S_i$$

Measures variation inside each class.

LDA: Between-Class Scatter Matrix

Global mean:

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i$$

Between-class scatter:

$$S_B = \sum_{i=1}^c N_i (\mu_i - \mu)(\mu_i - \mu)^T$$

where:

- N_i = samples in class i
- μ_i = class mean

Measures separation among classes.

LDA seeks projection vector w that maximizes

$$J(w) = \frac{w^T S_B w}{w^T S_W w}$$

This is known as the Fisher Criterion.

Solution:

$$S_B w = \lambda S_W w$$

which is a generalized eigenvalue problem.

Select eigenvectors with largest eigenvalues.

LDA Optimization: Fisher Criterion

$$J(w) = \frac{w^T S_B w}{w^T S_W w}, \quad w^T S_W w = 1$$

$$L(w, \lambda) = w^T S_B w - \lambda(w^T S_W w - 1)$$

$$\frac{\partial L}{\partial w} = 0 \Rightarrow 2S_B w - 2\lambda S_W w = 0$$

$$S_B w = \lambda S_W w$$

$$S_W^{-1} S_B w = \lambda w$$

Eigenvectors of $S_W^{-1} S_B \rightarrow$ LDA directions

LDA: Simple Example (Two-Class Case)

Goal: Find a direction that best separates two classes.

Given:

- Class 1 mean: μ_1
- Class 2 mean: μ_2
- Within-class scatter: S_W

Fisher LDA Projection:

$$w = S_W^{-1}(\mu_1 - \mu_2)$$

Step 1: Compute class means

$$\mu_1 = \frac{1}{n_1} \sum_{x \in C_1} x, \quad \mu_2 = \frac{1}{n_2} \sum_{x \in C_2} x$$

Step 2: Project data

$$y = w^T x$$

LDA Numerical Example: 2D to 1D Projection

Given data:

$$C_1 : (1, 2), (2, 1) \quad C_2 : (4, 5), (5, 4)$$

Step 1: Class means

$$\mu_1 = \left(\frac{1+2}{2}, \frac{2+1}{2} \right) = (1.5, 1.5)$$

$$\mu_2 = \left(\frac{4+5}{2}, \frac{5+4}{2} \right) = (4.5, 4.5)$$

Step 2: LDA direction

$$w \propto (\mu_2 - \mu_1)$$

$$w = (4.5 - 1.5, 4.5 - 1.5) = (3, 3)$$

Normalize:

$$w = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

Step 3: Projection formula

$$y = w^T x = \frac{x_1 + x_2}{\sqrt{2}}$$

Step 4: Projected values

$$(1, 2) \rightarrow \frac{3}{\sqrt{2}} \approx 2.12$$

$$(2, 1) \rightarrow \frac{3}{\sqrt{2}} \approx 2.12$$

$$(4, 5) \rightarrow \frac{9}{\sqrt{2}} \approx 6.36$$

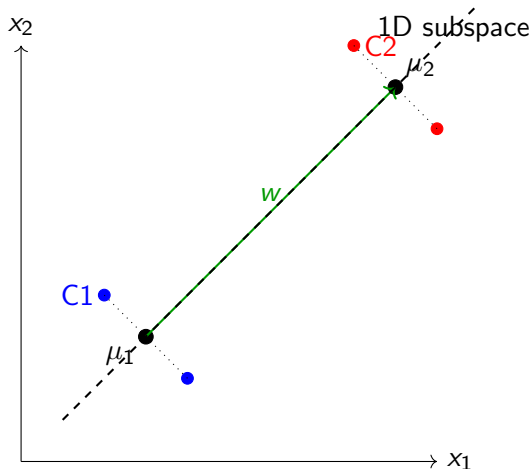
$$(5, 4) \rightarrow \frac{9}{\sqrt{2}} \approx 6.36$$

Final result:

$$C_1 \rightarrow \{2.12, 2.12\}, \quad C_2 \rightarrow \{6.36, 6.36\}$$

Conclusion: 2D data is reduced to 1D while preserving class separability.

LDA: 2D to 1D Projection (Visualization)



Idea: Project all points onto direction $w = (\mu_2 - \mu_1)$ to get 1D representation.

Visualization of High-Dimensional Data

After applying dimensionality reduction techniques such as PCA or LDA, high-dimensional data

$$X \in \mathbb{R}^{n \times d}$$

is mapped into a low-dimensional embedding

$$Z \in \mathbb{R}^{n \times 2} \quad \text{or} \quad Z \in \mathbb{R}^{n \times 3}$$

which can be visualized as scatter plots.

Why visualization is important?

- Understand global structure of data
- Identify clusters and class separability
- Detect outliers and noise
- Evaluate effectiveness of feature extraction

t-SNE (t-Distributed Stochastic Neighbor Embedding)

- Preserves local neighborhood structure
- Converts similarities into probability distributions
- Excellent for cluster visualization

UMAP (Uniform Manifold Approximation and Projection)

- Preserves local and partial global structure
- Based on manifold learning and graph theory
- Faster and more scalable than t-SNE

Key Idea:

High-dimensional data $\rightarrow$ 2D/3D embedding for visualization

Comparison of Dimensionality Reduction Methods

Method	Type	Goal	
PCA	Unsupervised	Maximize variance	
LDA	Supervised	Maximize class separability	
t-SNE	Unsupervised	Local structure preservation	P
UMAP	Unsupervised (can be supervised)	Local + partial global structure	

Key Insight:

PCA (variance) $\rightarrow$ LDA (separation) $\rightarrow$ t-SNE / UMAP (visualization)